TUGAS MOBILE - 2

PEMROGRAMAN MOBILE

KELOMPOK 1

AKFINA NI'MAWATI

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MENU

LOGIN PAGE

CODE

LOGIN PAGE

User memasukkan username dan password melalui TextField.

```
final TextEditingController _usernameController = TextEditingController();
final TextEditingController _passwordController = TextEditingController();
```

Ambil nilai input

```
String username = _usernameController.text.trim();
String password = _passwordController.text.trim();
```

Validasi

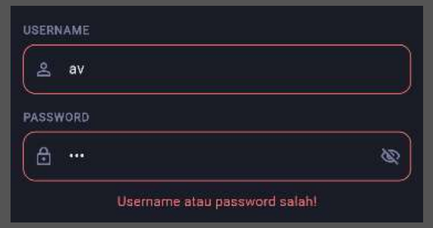
```
if (userList.any((u) => u.username == username && u.password ==
password)) {
```

Login Berhasil

```
isLoggedin = true;
Navigator.pushReplacement(
  context,
  MaterialPageRoute(builder: (context) => const HomeScreen()),);
```

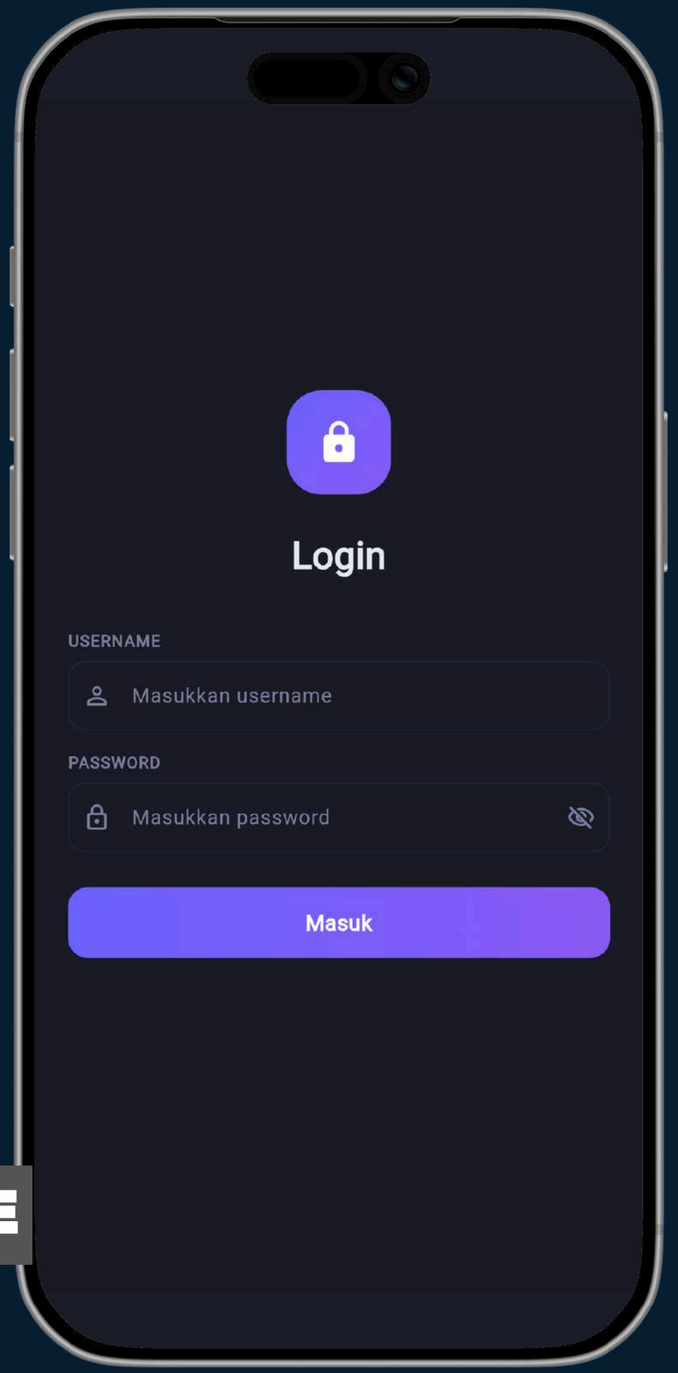
Login Gagal

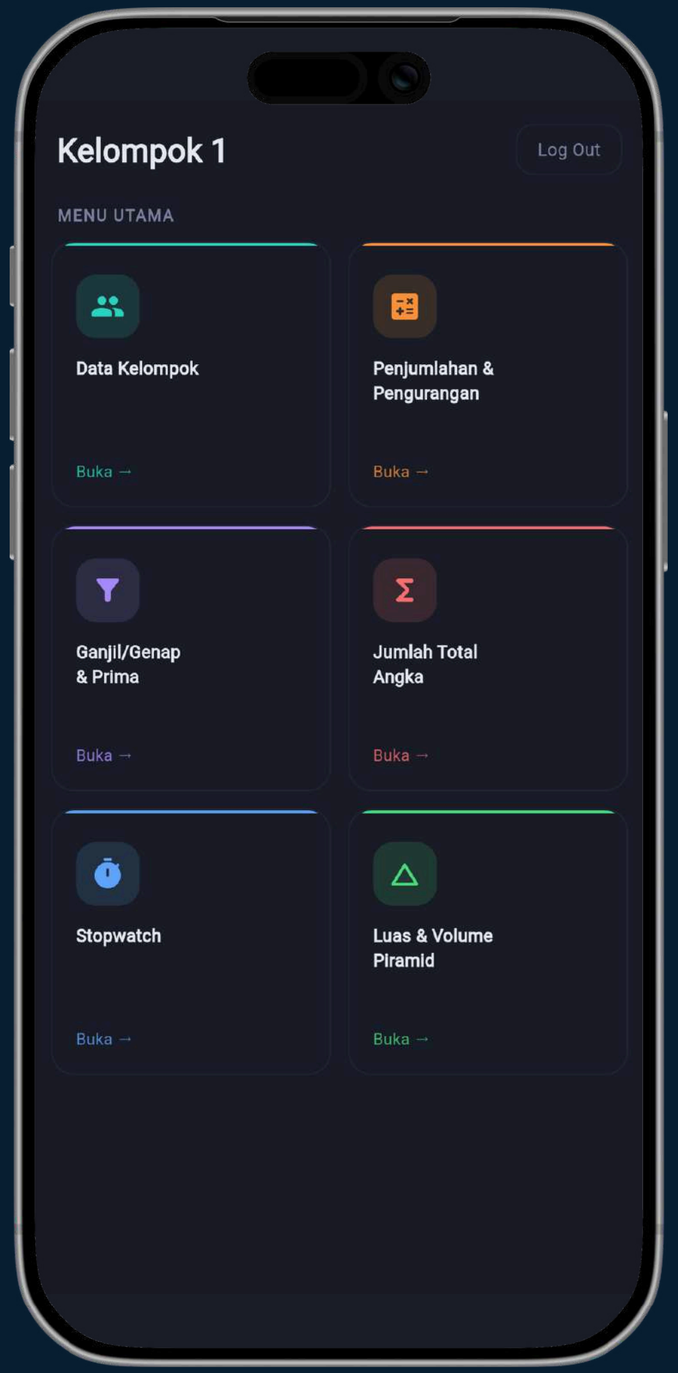
```
isLoginFailed = true;
```

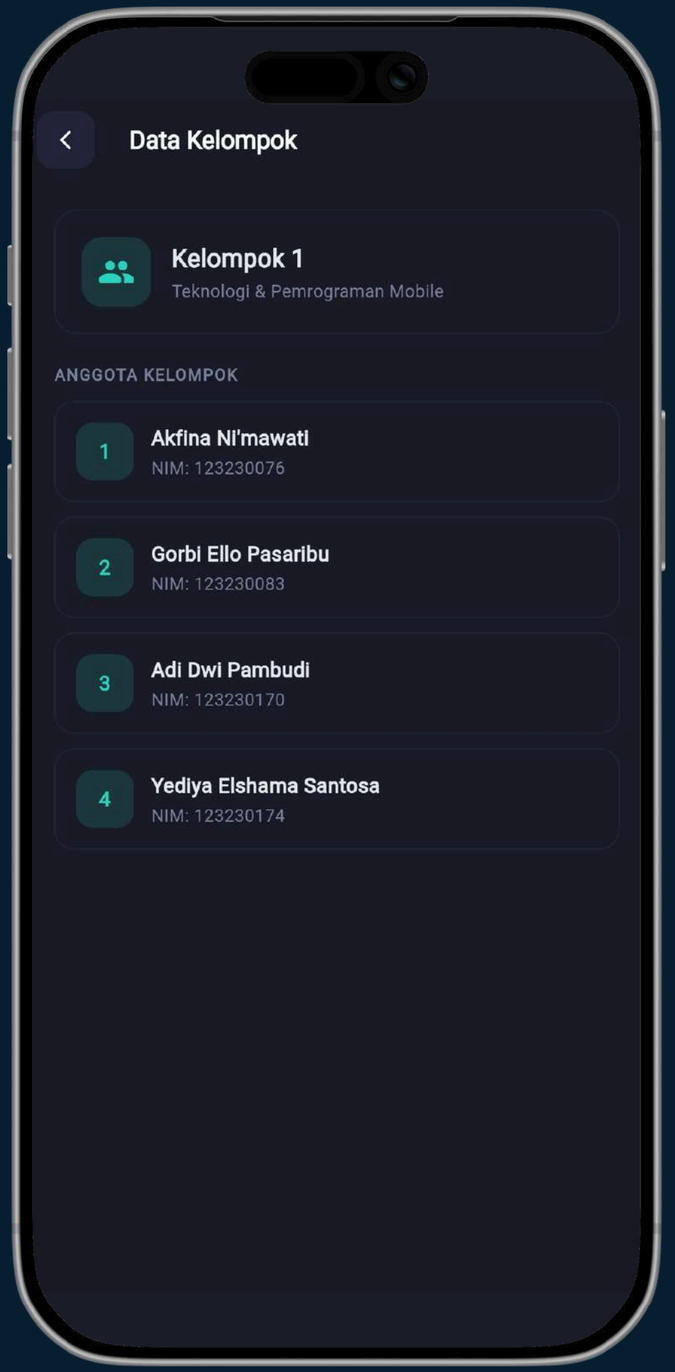


OUTPUT

LOGIN PAGE







MENU

PENJUMLAHAN & PENGURANGAN

User input angka (a = angka pertama; b = angka kedua)

```
final _aCtrl = TextEditingController();
final _bCtrl = TextEditingController();
```

Fungsi penghitungan

```
void _hitung(String op) {
  final a = double.tryParse(_aCtrl.text);
  final b = double.tryParse(_bCtrl.text);
```

Validasi

```
if (a == null || b == null) {
  _hasil = 'error';  return;}
```

Operasi

```
final result = op == '+' ? a + b : a - b;
```

Hasil

```
setState(() {
  _operasi = op;
  _hasil = result.toString();
});
```

CODE

PENJUMLAHAN & PENGURANGAN

OUTPUT

PENJUMLAHAN & PENGURANGAN

MENU

GANJIL, GENAP, PRIMA

CODE

GANJIL, GENAP, PRIMA

Input bilangan

```
final _ctrl = TextEditingController();
```

Cek Ganjil Genap

```
bool isGanjil = n % 2 != 0;
```

Cek Prima

```
bool isPrima(int n) {
  if (n < 2) return false;
  for (int i = 2; i <= n ~/ 2; i++) {
    if (n % i == 0) return false;
  }
  return true;
}
```

Input bilangan **n**
Jika n % 2 = 0 → Genap
Jika n % 2 ≠ 0 → Ganjil

Jika n < 2 → Bukan prima
Untuk i = 2 sampai n/2
    Jika n % i = 0 → Bukan prima
Jika tidak ada pembagi → Prima

OUTPUT

GANJIL, GENAP, PRIMA

MENU

JUMLAH TOTAL ANGKA

Menyimpan data angka

```
final List<double> _list = [];
double _total = 0;
```

Validasi

```
if (val == null) {
  _error = 'Masukkan angka yang valid!';
  return;
}
```

Tambah list dan penghitungan total angka

```
_list.add(val);
_total += val;
```

Hapus angka

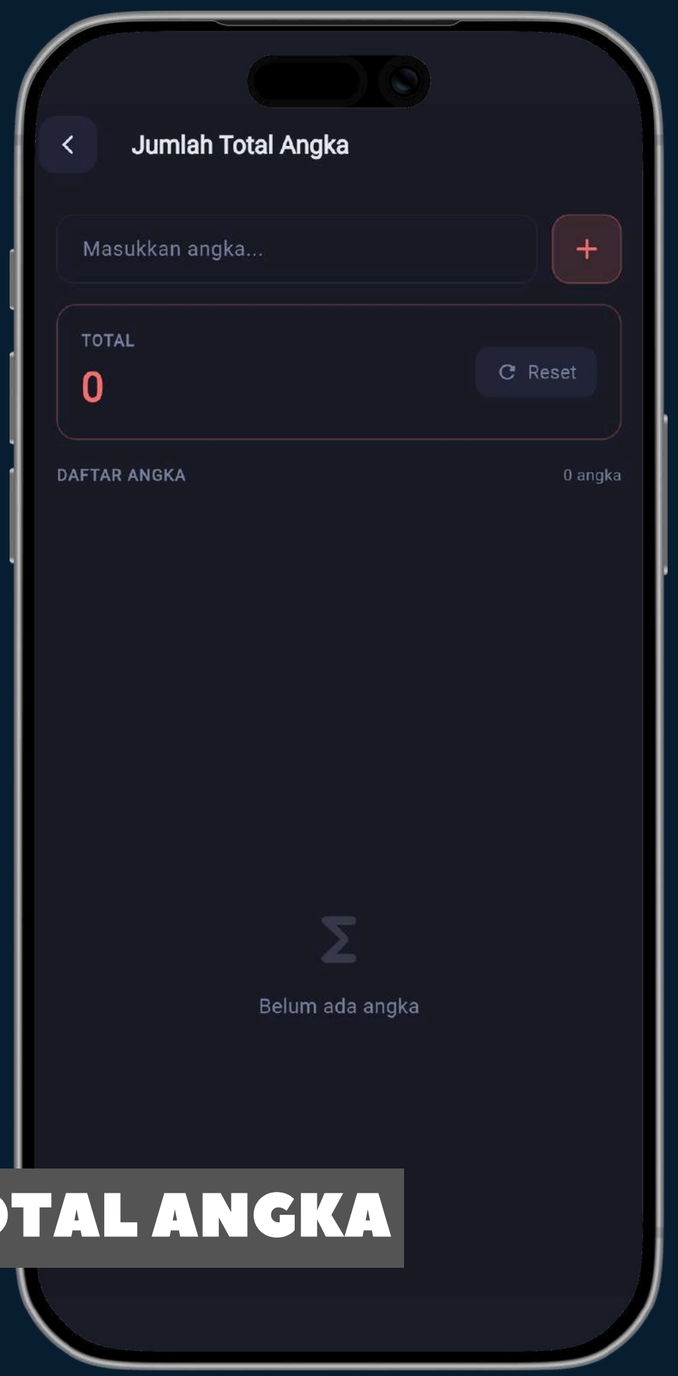
```
_total -= _list[index];
_list.removeAt(index);
```

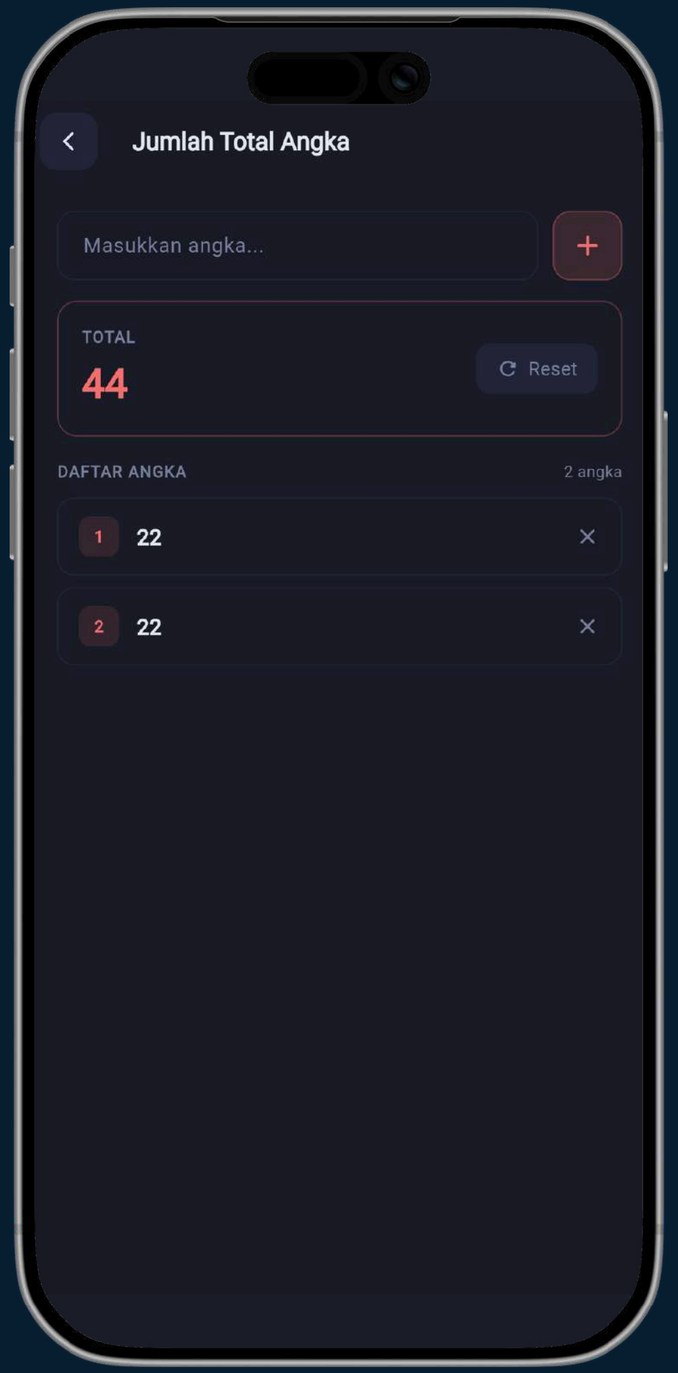
Reset data

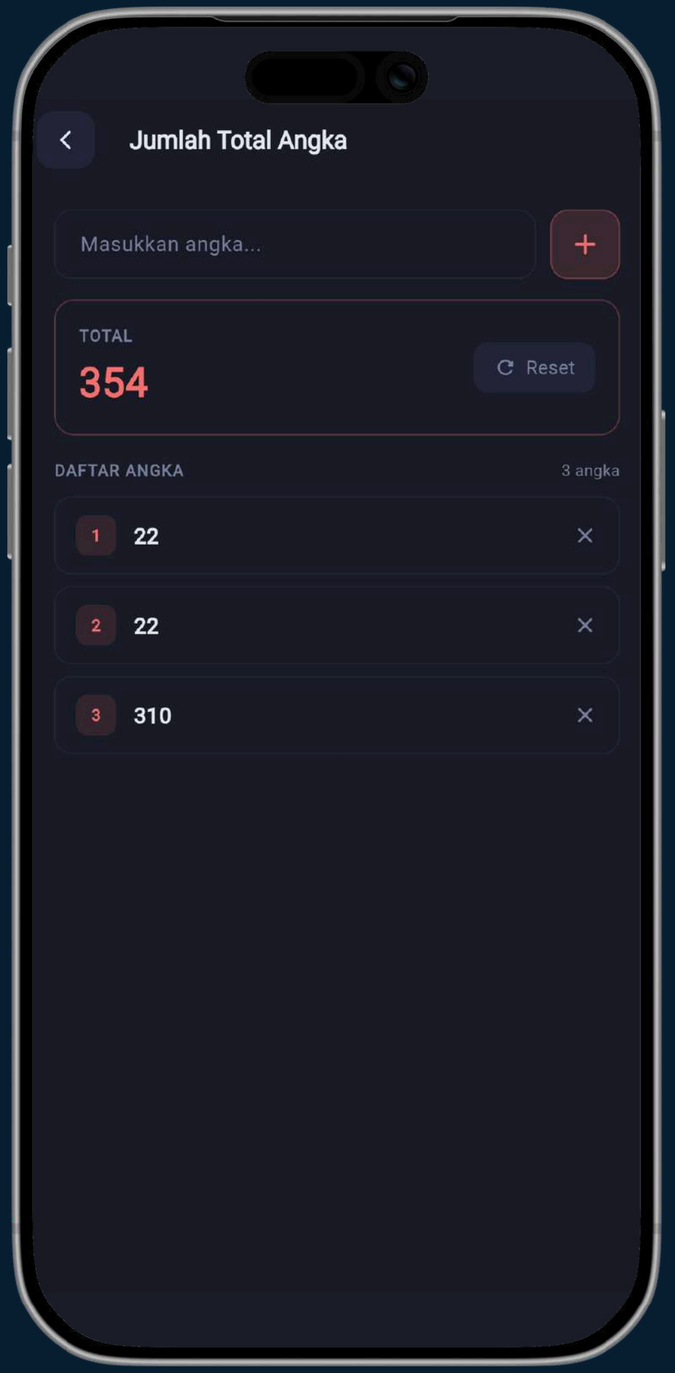
```
_list.clear();
_total = 0;
```

CODE

JUMLAH TOTAL ANGKA







OUTPUT

JUMLAH TOTAL ANGKA

MENU

STOPWATCH

CODE

STOPWATCH

Variabel

```
Timer? _timer;
int _ms = 0;
bool _running = false;
final List<String> _laps = [];
```

Memulai stopwatch

```
void _start() {
  _timer = Timer.periodic(Duration(milliseconds: 10), (_) {
    _ms++;
  });
  _running = true;
}
```

Stop stopwatch

```
void _stop() {
  _timer?.cancel();
  _running = false;
}
```

Menyimpan lap

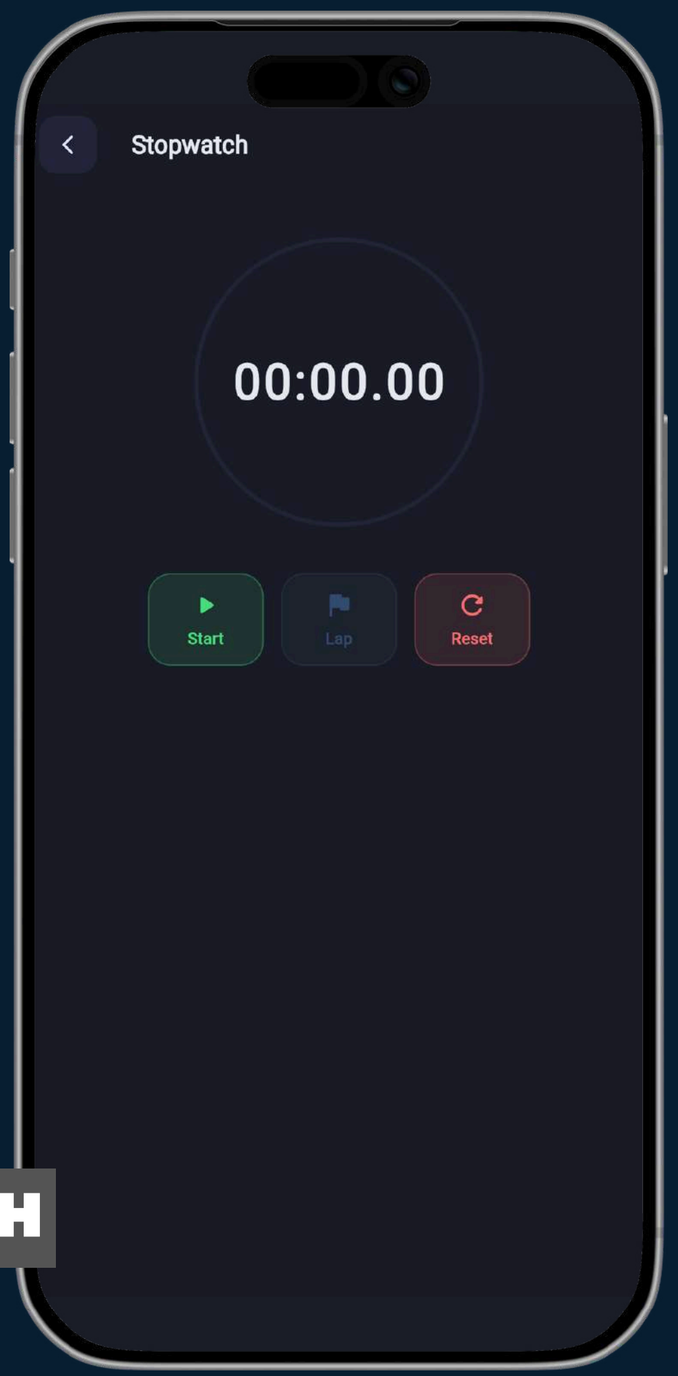
```
void _lap() {
  _laps.insert(0, 'Lap ${_laps.length + 1}
$_time');
}
```

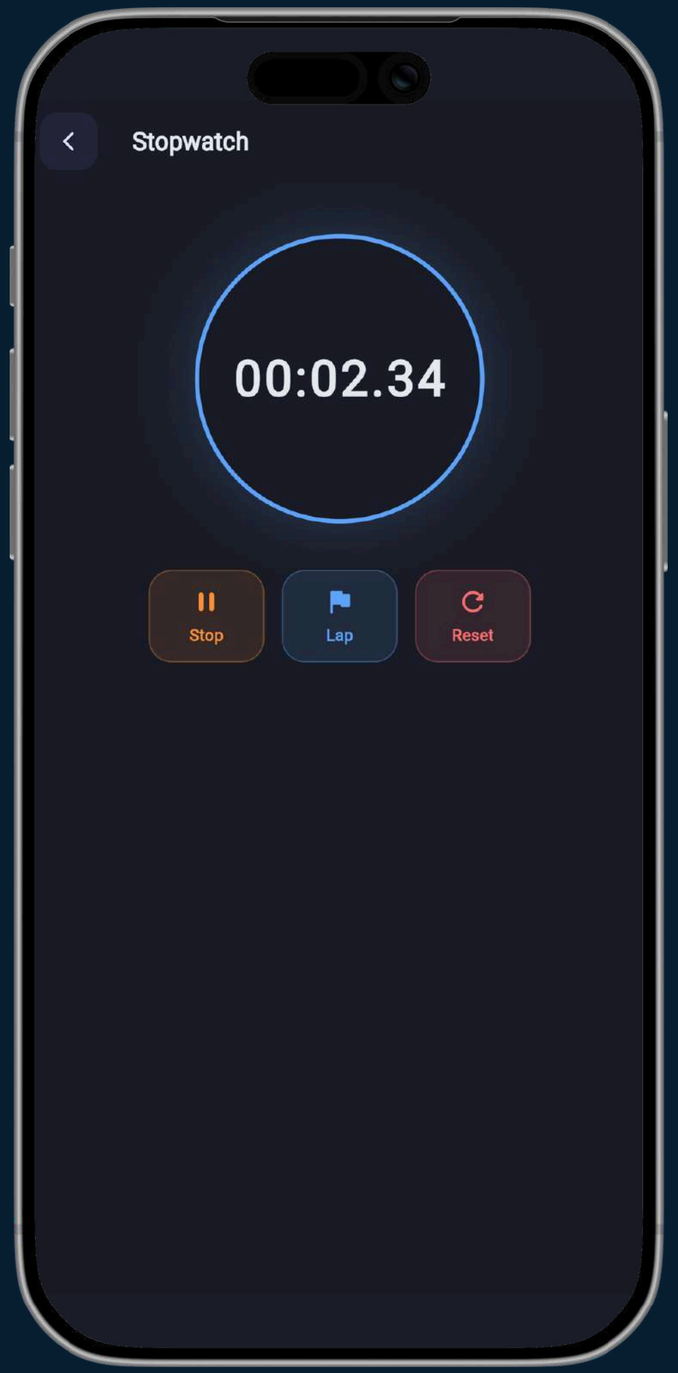
Format waktu

```
final ms = _ms % 100;
final sec = (_ms ~/ 100) % 60;
final min = _ms ~/ 6000;
```

OUTPUT

STOPWATCH





MENU

PIRAMID

Input data

```
final alas = double.tryParse(_alasCtrl.text.trim());
final tinggi = double.tryParse(_tinggiCtrl.text.trim());
```

Validasi

```
if (alas == null || tinggi == null || alas <= 0 || tinggi <= 0)
```

Luas alas

luasAlas = alas × alas

Apotema

$apotema = \sqrt{tinggi^2 + (alas/2)^2}$

Apotema

$apotema = \sqrt{tinggi^2 + (alas/2)^2}$

Luas selimut

$luasSelimut = 4 \times \frac{1}{2} \times alas \times apotema$

Luas permukaan

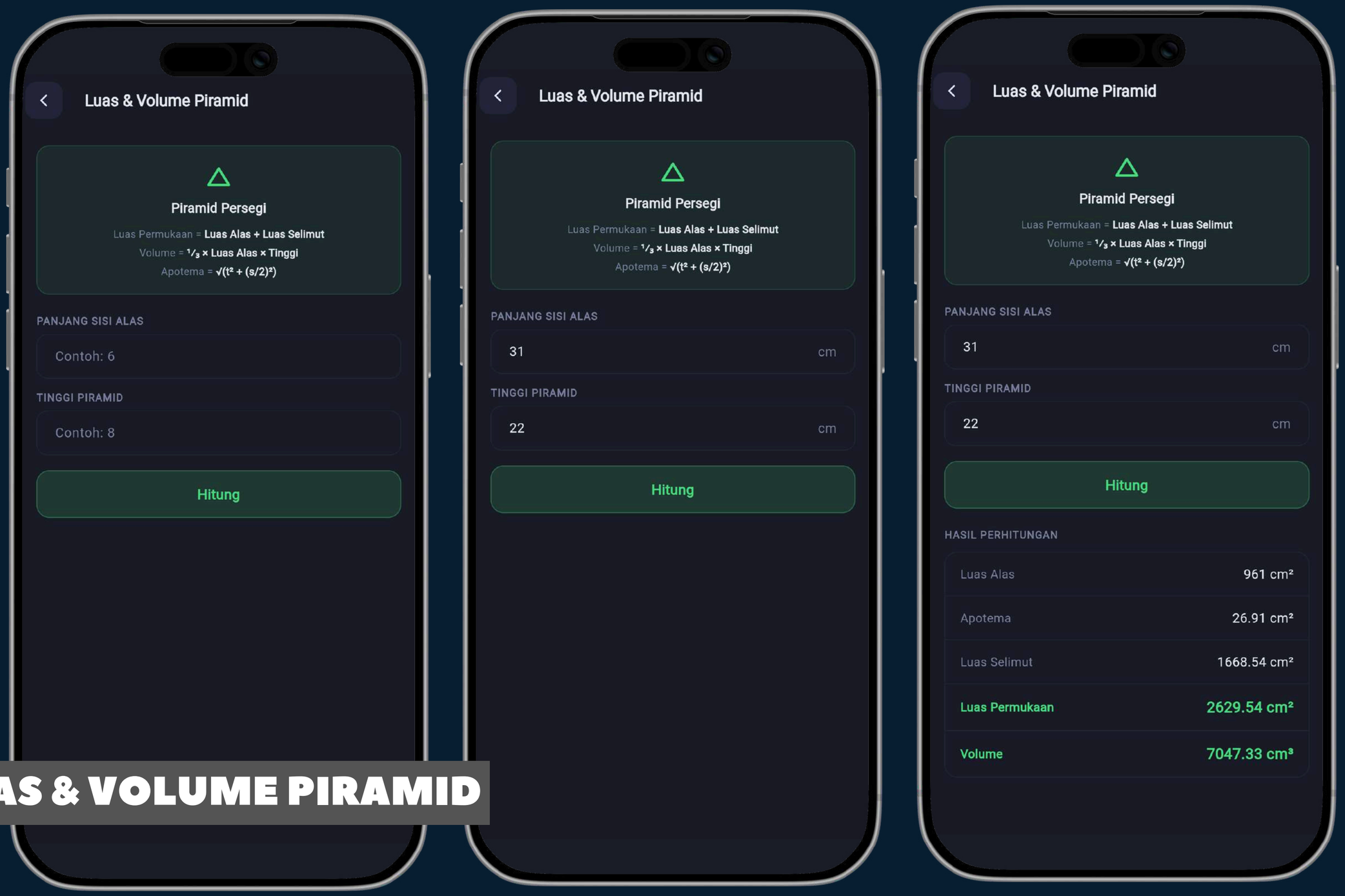
luasPermukaan = luasAlas + luasSelimut

Volume

$volume = \frac{1}{3} \times luasAlas \times tinggi$

CODE

PIRAMID



OUTPUT

HITUNG LUAS & VOLUME PIRAMID

THANK YOU

PEMROGRAMAN MOBILE